

Outline (until lunch break)

- Hands-on: Association of Barcode/Tag Sequences
- Hands-on: Count Sequencing Analysis
- Discussion of QC metrics
- Hands-on: Data Analysis Steps (Regions and Variants)

Hands-on: Association of Barcode/Tag Sequences



ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

Max Schubach
Computational Genome Biology @ BIH

July 20, 2025



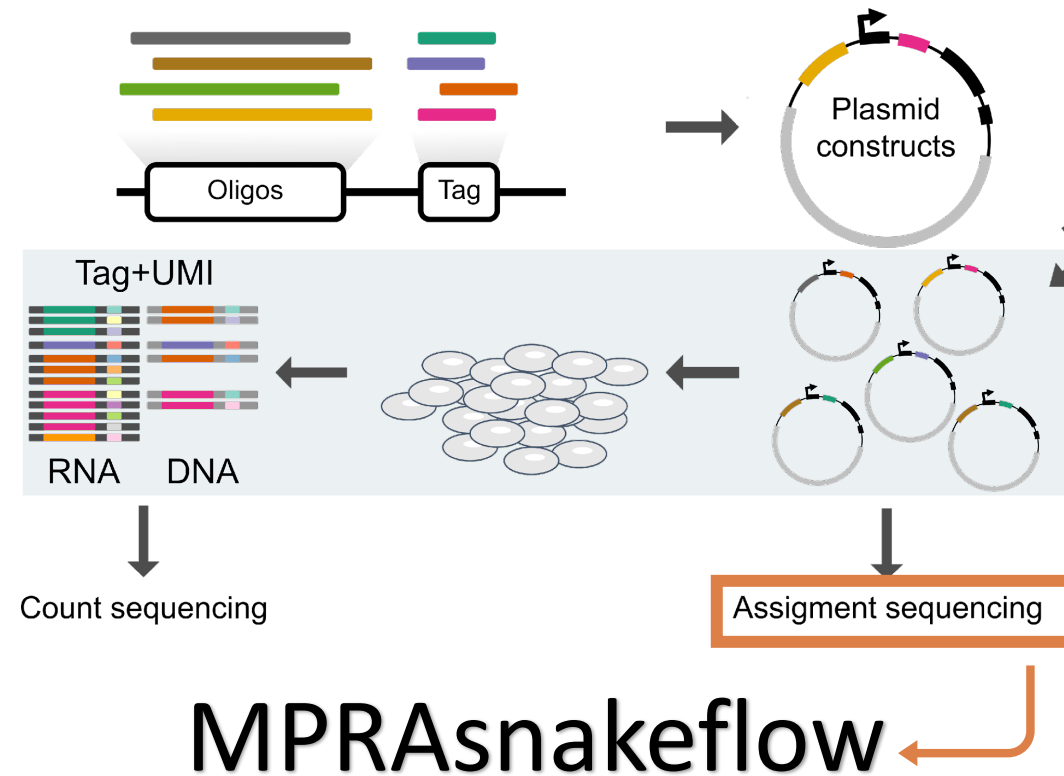
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MPRAsnakeflow – Pipeline to process sequencing data from MPRA to create count tables for candidate sequences and variants



Running the association workflow



03 Standard Pipeline for MPRA Analysis: MPRAsnakeflow

External links:

- [MPRAsnakeflow repository](#)
- [MPRAsnakeflow documentation](#)
- [MPRAsnakeflow tutorial](#)

Hands-on:

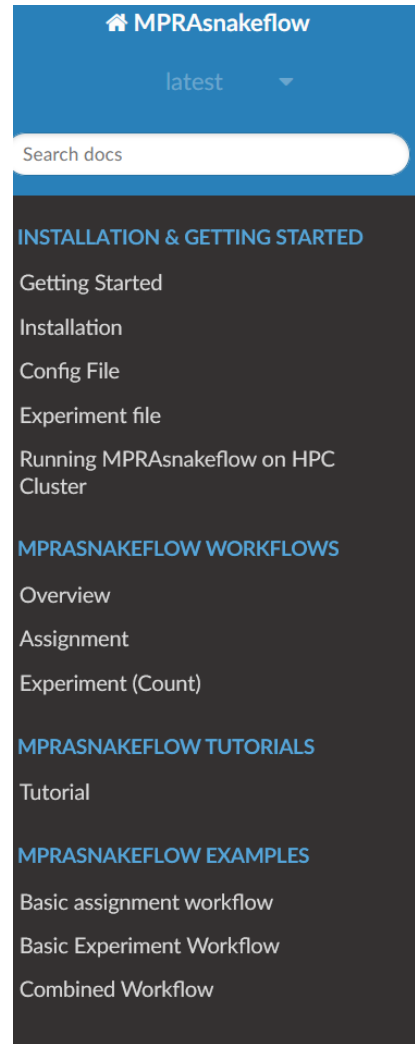
- Association of Barcode/Tag Sequences: [Jupyter/Colab notebook](#)
- Count Sequencing Analysis: [Jupyter/Colab notebook](#)



https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

MPRAsnakeflow

<https://github.com/kircherlab/MPRAsnakeflow>



[MPRAsnakeflow's Documentation](#)

[View page source](#)

MPRAsnakeflow's Documentation

snakemake `≥8.24.1` conda `>24.7.1`

Welcome!

MPRAsnakeflow is a pipeline designed to process sequencing data from Massively Parallel Reporter Assays (MPRAs) to create count tables for candidate sequences tested in the experiment.

MPRAsnakeflow is built on top of [Snakemake](#) (version `≥8.24.1` required) and is configured via a `.yaml` file.

Authors

- Max Schubach ([@visze](#)) | [Computational Genome Biology Group](#) | Berlin Institute of Health at Charité - Universitätsklinikum Berlin

Usage

If you use this workflow in a paper, don't forget to give credit to the authors by citing the URL of the (original) repository.

Installation & Getting Started

<https://mprasnakeflow.readthedocs.io>

Technical details

- **MPRAsnakeflow is implemented in snakemake**
 - Python based workflow management software
 - <https://snakemake.readthedocs.io>
- **Software packages managed via conda or apptainer**
- **Input controlled via config files (YAML)**

Possible Snakemake commands

```
# basic command
snakemake --sdm conda --cores 4 --configfile config.yaml

# run within working dir
snakemake --sdm conda -c 4 --configfile config.yaml \
--snakefile path/to/MPRAsnakeflow/workflow/Snakefile

# use different working dir (run in MPRAsnakeflow)
snakemake --sdm conda -c 4 --configfile config.yaml \
--directory path/to/working/dir
```

Config file example

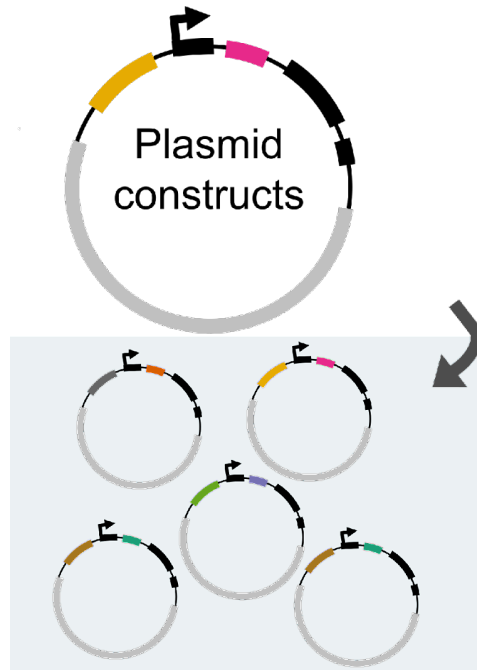
```
assignments:
  exampleAssignment: # name of your analysis
    bc_length: 15
    alignment_tool:
      tool: bbmap
      configs:
        sequence_length: 270
        alignment_start: 1
    FW: [resources/Assignment_Basic/R1.fastq.gz]
    BC: [resources/Assignment_Basic/R2.fastq.gz]
    REV: [resources/Assignment_Basic/R3.fastq.gz]
    design_file: resources/design.fa
    configs:
      default: {} # name of additional filtering
```

MPRAsnakeflow

How?

- Assignment → assign barcodes to corresponding tested elements (optional)
- Experiment → process sequenced barcodes to create count tables

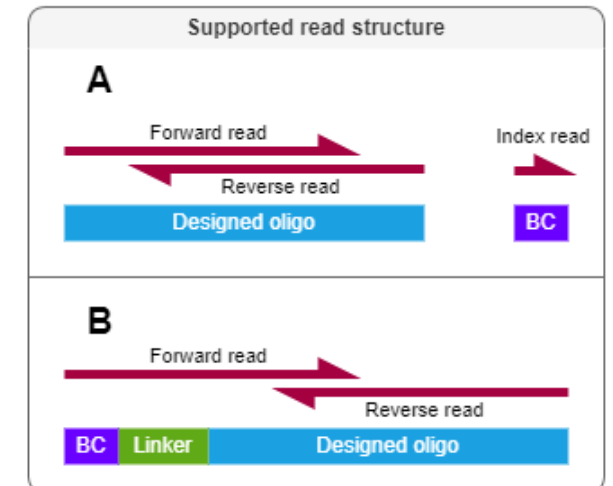
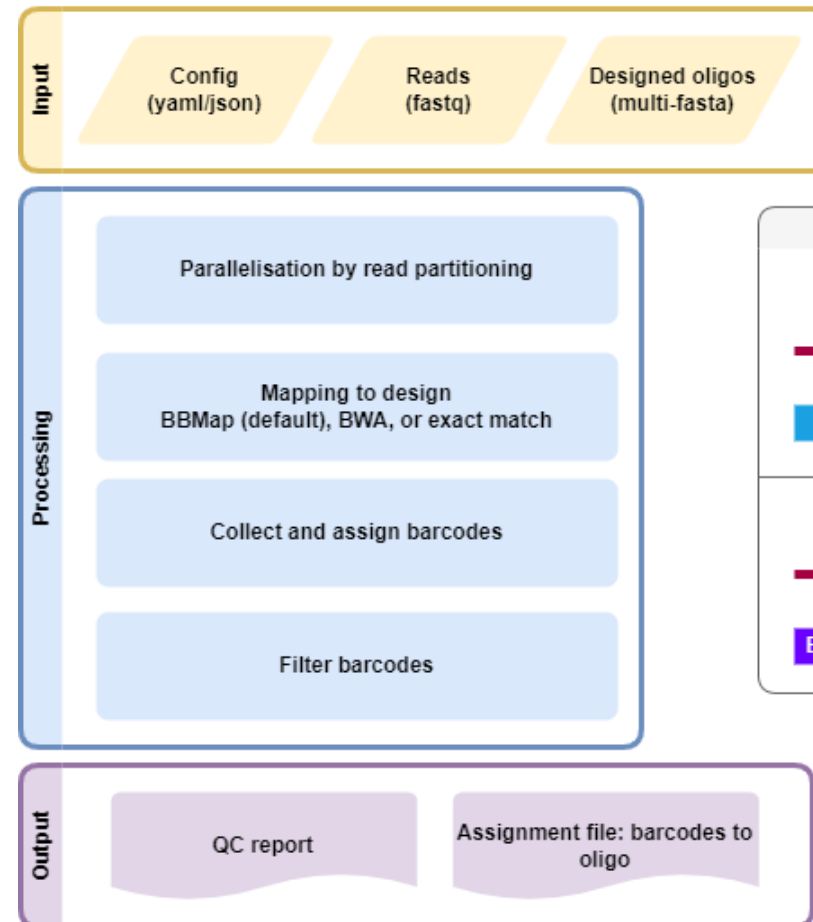
Assignment Workflow: Overview



Assignment sequencing

- (PE) sequencing reads of barcode
- Sample index and/or UMI reads

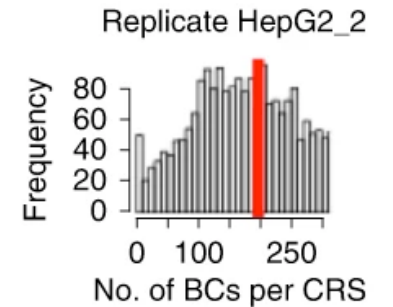
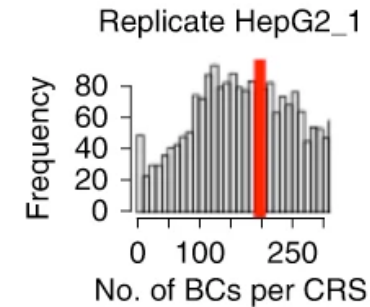
Assignment workflow



MPRAsnakeflow: assignment workflow

Why?

- Require correct assignment of barcodes to sequences
 - Mis-assignment could lead to misinterpretation of element activity, or worse, skewing of a variant log ratio
- Look for:
 - Sufficient coverage of BC / seq
 - Avoid BC mapping to multiple sequences
 - Measure library complexity and achieve close to complete coverage in association sequencing



Running the association workflow



03 Standard Pipeline for MPRA Analysis: MPRAsnakeflow

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Hands-on:

- Association of Barcode/Tag Sequences: [Jupyter/Colab notebook](#)
- Count Sequencing Analysis: [Jupyter/Colab notebook](#)



https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

Thank You! Questions?

kircherlab.github.io

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Hands-on: Count Sequencing Analysis



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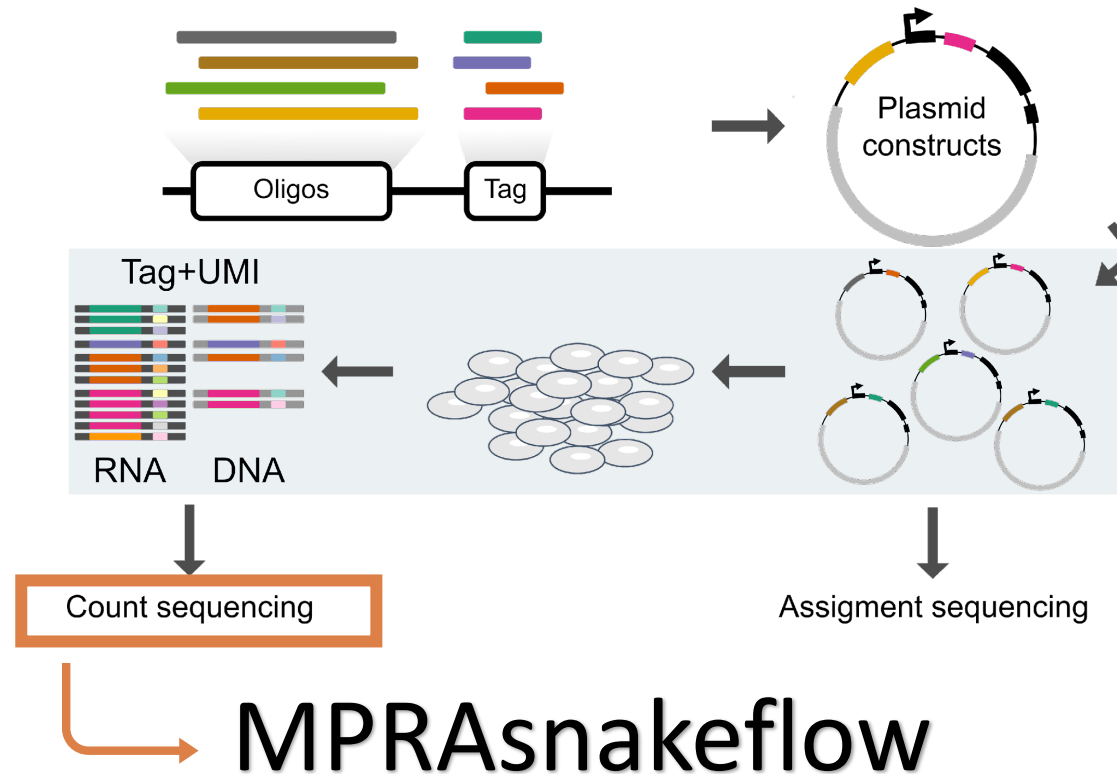
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Running the count/experiment workflow



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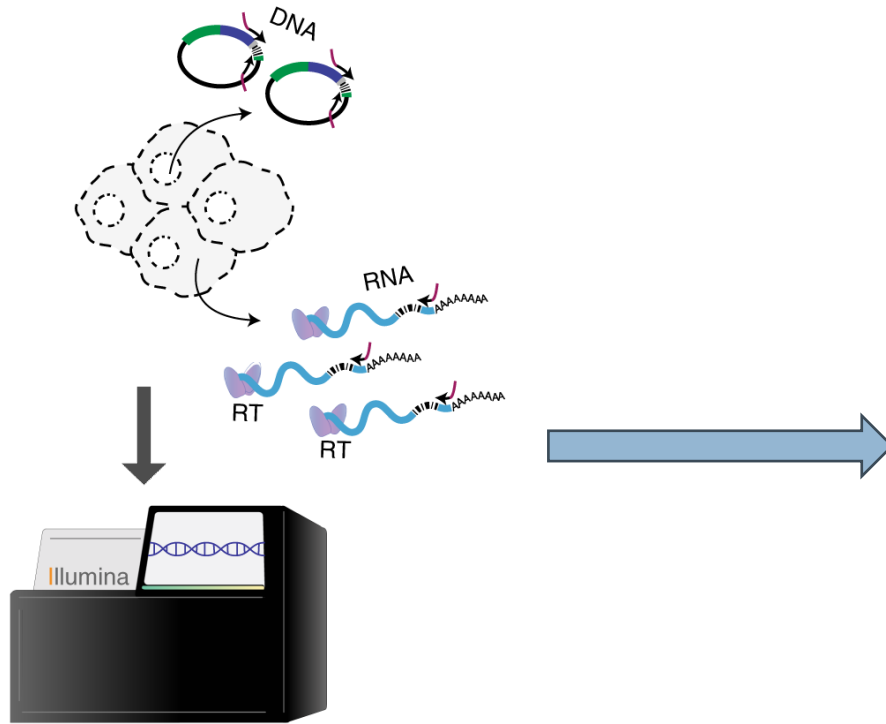
Hands-on:

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- Count Sequencing Analysis: [Jupyter/Colab notebook](#)



https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

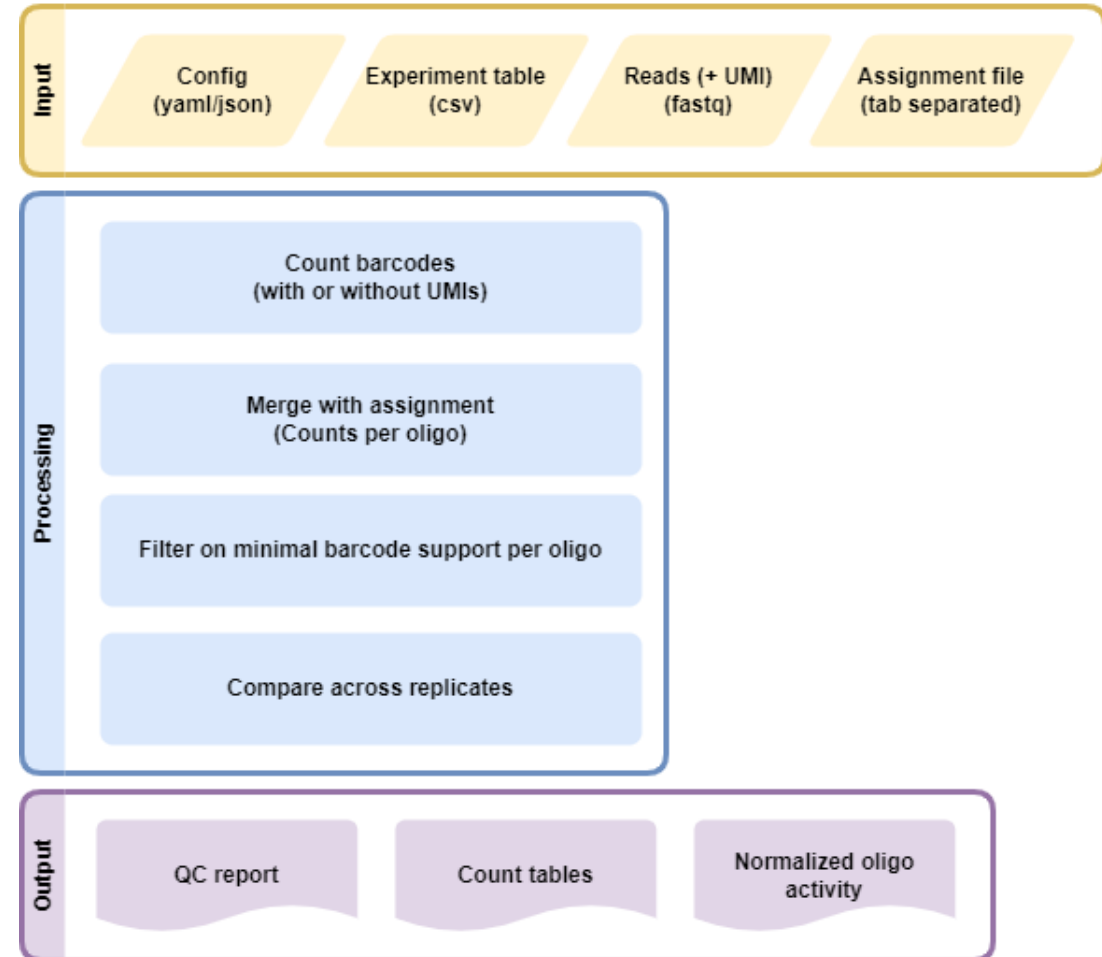
Experiment Workflow: Overview



Assignment sequencing

- Sequencing reads of plasmid insert
- Target barcode
- Sample index

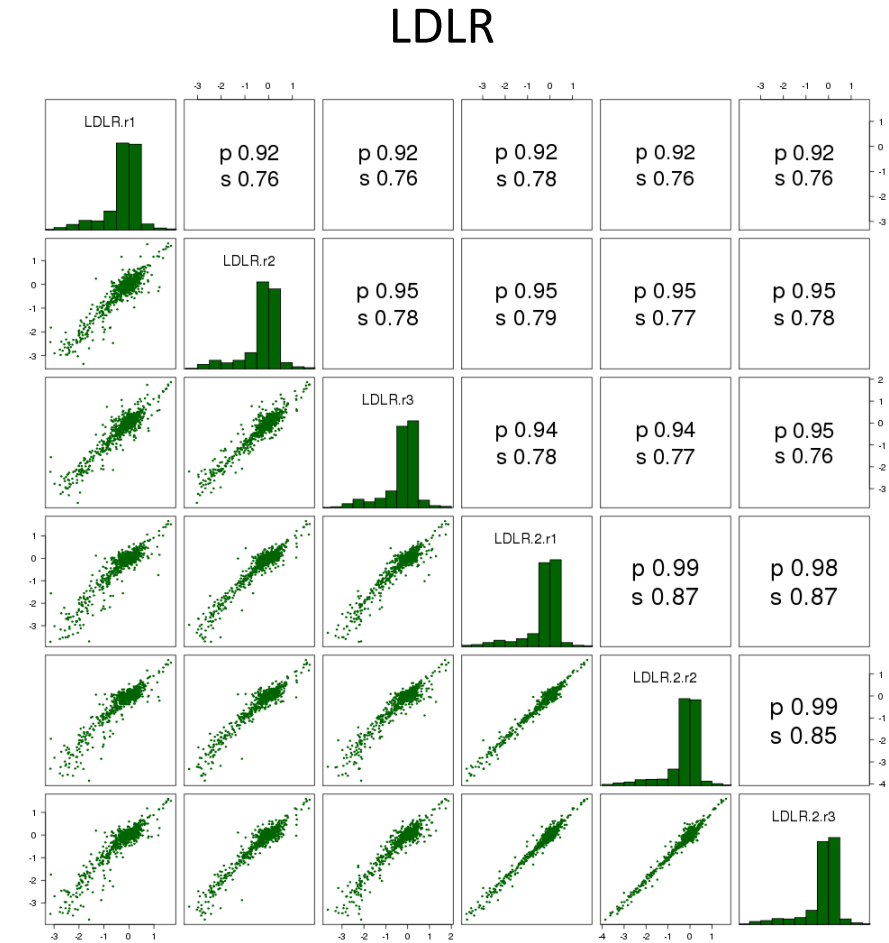
Experiment workflow



MPRAsnakeflow: Experiment Workflow

Why?

- Need to summarize and integrate data
 - Each tested element is represented by many barcodes – information stored in the assignment
 - Barcode sequence might influence measured activity
 - Variants are represented by at least two elements
- Look for:
 - Sufficient coverage of elements with barcodes
 - Good correlation between replicates



Combined replicates: p 0.97 / s 0.84

<https://www.nature.com/articles/s41467-019-11526-w>

(Saturation mutagenesis data)

Running the count workflow



03 Standard Pipeline for MPRA Analysis: MPRAsnakeflow

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Hands-on:

- Association of Barcode/Tag Sequences: [Jupyter/Colab notebook](#)
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Discussion of QC metrics

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Discussion of QC metrics

1. QC report assignment workflow:

- https://htmlpreview.github.io/?https://github.com/kircherlab/IS-MB-2025-IGVF-MPRA-Tutorial/blob/main/03_MPRAsnakeflow/qc_report.assignment.html

2. QC report experiment workflow:

- https://htmlpreview.github.io/?https://github.com/kircherlab/IS-MB-2025-IGVF-MPRA-Tutorial/blob/main/03_MPRAsnakeflow/qc_report.experiment.html

MPRAsnakeflow assignment QC report

Code Table of contents
Overview
Overall quality metrics
BC distribution over oligos
Counts of barcodes and oligos

Overview

The Assignment QC (Quality Control) Report is a comprehensive document designed to evaluate and ensure the quality of data or processes involved in a specific assignment. This report is generated for the assocBasic assignment with settings defined in the config default. The barcode length for this experiment is 15 bases.

- **Forward:** data/SRR10800986_1.fastq.gz
- **Reverse:** data/SRR10800986_3.fastq.gz
- **Barcode:** data/SRR10800986_2.fastq.gz
- **Design File:** design.fa

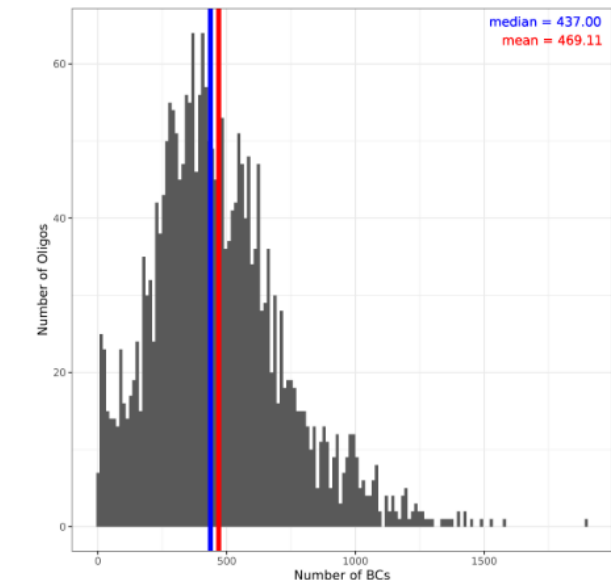
Overall quality metrics

Table explanation

- **median assigned barcodes:** Median number of barcodes assigned to tested sequences in mapping as a quality control measure for the assignment step, whether there is sufficient barcode to oligo coverage.
- **fraction assigned oligos:** Fraction of assigned tested sequences in mapping to determine if the library during the assignment step was sufficiently recovered.

median assigned barcodes	fraction assigned oligos
437	1.00

BC distribution over oligos



Thank You! Questions?

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MPRA analysis with BCalm

ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

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Hands-on: Identifying regions and variants with significant effects



04 MPRA Data Analysis

Hands-on:

- Data Analysis Steps (Regions and Variants): [README](#), [Jupyter/Colab notebook](#)

https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

What are we looking for?

Element Analysis: *"Is this sequence regulatory active?"*

Variant Analysis: *"Does this variant change regulatory activity?"*

Element 1	...ATCGC...		...ATCGC...		Reference
Element 2	...CGTCC...		...AT T GC...		Variant 1
Element 3	...CAGCT...		...AT A GC...		Variant 2
Neg Control	...TGCAA...		...AT G GC...		Variant 3
<i>Expression</i>			<i>Expression</i>		

Revisiting the data structure from MPRA experiment

Barcode level:

- Each test element represented by many barcodes
- High variance of the RNA/DNA ratio per barcode
- Barcode sequence might influence activity

Oligo level:

- Robust activity measurement for oligos
- Outlier barcodes may indicate sequence-specific effects?

Replicate level:

- Technical replicates for experimental reproducibility
- Replicate sample normalized for read depth

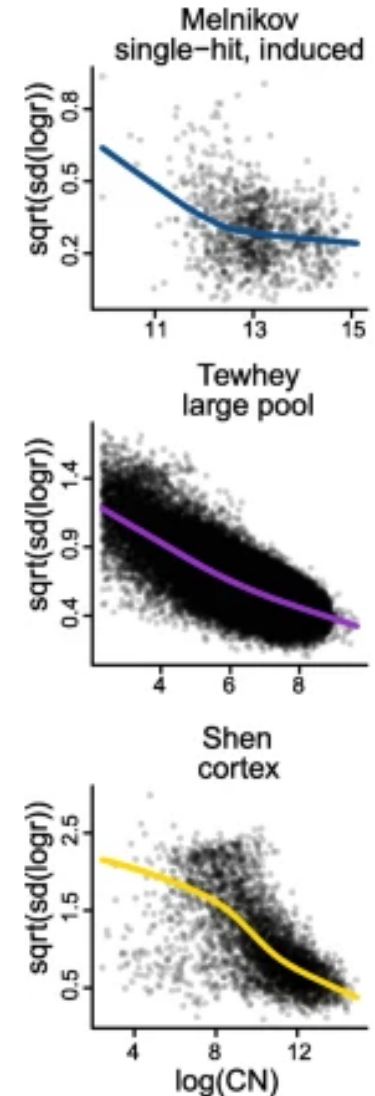
Barcode	name	dna_count_1	rna_count_1
AAAAAAAAAAGCTGC	oligo_006364	11	10
AAAAAAAAATCACAG	oligo_005641	8	7
AAAAAAAAATTGAGC	oligo_005719	2	25
AAAAAAAAATTGTAT	oligo_005725	4	12
AAAAAAAACACATGA	oligo_005412	21	15
AAAAAAAACATAGTT	oligo_006471	8	3

Revisiting the data structure from MPRA experiment

Counts follow a distribution where the variance increases with the mean

Elements with lower DNA counts have much more variable activity measurements

Different approaches have different statistical properties - average estimator OR aggregate estimator?



mpralm - linear models to analyze MPRA data

General tools repurposed for MPRA analysis

- DESeq2 for significant transcriptional activity
- t-test for significant differential activity

MPRA analysis specific tools

- MPRAnalyze
- mpralm
- BCalm

But why?

voom

mpralm

empirical Bayes

Method | [Open access](#) | Published: 02 September 2019

MPRAnalyze: statistical framework for massively parallel reporter assays

[Tal Ashuach](#), [David S. Fischer](#), [Anat Kreimer](#), [Nadav Ahituv](#), [Fabian J. Theis](#) & [Nir Yosef](#) 

[Genome Biology](#) **20**, Article number: 183 (2019) | [Cite this article](#)

Methodology article | [Open access](#) | Published: 12 March 2019

Linear models enable powerful differential activity analysis in massively parallel reporter assays

[Leslie Myint](#), [Dimitrios G. Avramopoulos](#), [Loyal A. Goff](#) & [Kasper D. Hansen](#) 

[BMC Genomics](#) **20**, Article number: 209 (2019) | [Cite this article](#)

Research | [Open access](#) | Published: 13 February 2025

Using individual barcodes to increase quantification power of massively parallel reporter assays

[Pia Keukeleire](#), [Jonathan D. Rosen](#), [Angelina Göbel-Knapp](#), [Kilian Salomon](#), [Max Schubach](#) & [Martin Kircher](#) 

BCalm - *mpralm*, but more robust against outliers

Problem with aggregation:

- When you sum up barcode counts, you lose information about variability between barcodes
- A single outlier barcode can heavily influence the aggregated result

BCalm's solution:

- Keep each barcode as a separate observation in the statistical model
- Gain statistical power from having more data points
- Robust handling of outlier barcodes through the modeling framework

Barcode	name	dna_count_1	rna_count_1
AAAAAAAAAAGCTGC	oligo_006364	11	10
AAAAAAAAATCACAG	oligo_005641	8	7
AAAAAAAAATTGAGC	oligo_005719	2	25
AAAAAAAAATTGTAT	oligo_005725	4	12
AAAAAACAACATGA	oligo_005412	21	15
AAAAAACAATAGTT	oligo_006471	8	3

And now we will ...

- Explore the dataset in detail
- Perform element analysis using BCalm
- Perform variant analysis using BCalm

Thank You!

kircherlab.github.io

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